GSSS INSTITUTE OF ENGINEERING & TECHNOLOGY FOR WOMEN, MYSURU
Department of Electronics & Communication Engineering | GATE Preparation Library

GATE Electronics & Communication Engineering (ECE) — Master Official Syllabus

Complete Prescribed Curriculum Covering All 10 Examination Sections with Marking Patterns

Total Subjects: 10 Subjects • **Total Marks:** 100 Marks (65 Questions) • **Examination Duration:** 3 Hours (180 Minutes) • **Organizing Body:** GATE Committee / IISc & IITs

GATE ECE OVERALL EXAMINATION STRUCTURE
Section 1: General Aptitude (GA): 10 Questions (5 of 1-mark + 5 of 2-marks) = **15 Marks**.
Section 2: Engineering Mathematics (EM): ~8-9 Questions = **13 Marks**.
Section 3 to 10: Core ECE Subjects: ~46-47 Questions = **72 Marks**.
Total: 65 Questions | 100 Marks | Negative Marking: 1/3 for 1-mark MCQ, 2/3 for 2-mark MCQ. Zero negative marking for MSQ and NAT.

ENGINEERING MATHEMATICS (13 Marks)

- **Linear Algebra:**
  Vector space, basis, linear dependence and independence of vectors.; Matrix algebra, algebra of symmetric, skew-symmetric, orthogonal, Hermitian, skew-Hermitian, and unitary matrices.; Eigenvalues and eigenvectors, algebraic and geometric multiplicity, characteristic equation, Cayley-Hamilton Theorem..
- **Calculus:**
  Mean value theorems: Rolle's theorem, Cauchy's mean value theorem, Lagrange's mean value theorem.; Theorems of integral calculus: Fundamental theorem of calculus, evaluation of definite and improper integrals.; Partial derivatives, total derivatives, chain rule, Euler's theorem on homogeneous functions..
- **Differential Equations:**
  First order equations (linear and non-linear): Integrating factor method, exact differential equations, Bernoulli's equation.; Higher order linear differential equations with constant coefficients: Complementary functions and particular integrals (shortcuts for exp, sin, cos, polynomials).; Cauchy-Euler differential equations and transformation to constant coefficient equations..
- **Complex Analysis:**
  Analytic functions, necessary and sufficient conditions for analyticity: Cauchy-Riemann equations in Cartesian and polar coordinates.; Harmonic functions and harmonic conjugates, orthogonal trajectories.; Contour integrals: Cauchy's integral theorem, Cauchy's integral formula for functions and derivatives..
- **Probability and Statistics:**
  Axiomatic definitions of probability, sampling theorems, conditional probability, independent events, Bayes' Theorem.; Random variables: Discrete and continuous random variables, Probability Mass Function (PMF), Probability Density Function (PDF), Cumulative Distribution Function (CDF).; Statistical parameters: Mean, median, mode, variance, standard deviation, moments, and moment generating functions..

Core References: Advanced Engineering Mathematics — Erwin Kreyszig (Wiley), Higher Engineering Mathematics — B.S. Grewal (Khanna Publishers)

NETWORK THEORY (8 - 11 Marks)

- **Circuit Analysis & Solution Methods:**

  Network solution methods: Nodal analysis, Mesh analysis, Supernode and Supermesh methods.; Node and mesh matrix equations: Inspection method for resistive and reactive networks with independent and dependent sources.; Network theorems: Superposition theorem (valid only for linear systems, power calculation trap), Thevenin's theorem, Norton's theorem..

- **Sinusoidal Steady-State Analysis & Resonance:**

  Sinusoidal steady-state analysis using phasors, impedance, and admittance concepts.; Active power (P), reactive power (Q), apparent power (S), and complex power. Power factor and power factor correction.; Resonance in electrical circuits: Series RLC resonance, parallel RLC resonance, anti-resonance..

- **Transient and Time-Domain Analysis:**

  Time-domain analysis of linear circuits: First-order RL and RC circuits, time constant calculation (tau = L/R, tau = RC).; Differential equations for second-order RLC circuits: Overdamped, critically damped, and underdamped transient responses.; Damping ratio (zeta) and undamped natural frequency (omega_n)..

- **Two-Port Networks & State Equations:**

  Two-port network parameter representations: Impedance parameters (Z), Admittance parameters (Y), Hybrid parameters (h), Inverse Hybrid parameters (g).; Transmission parameters (ABCD) and Inverse Transmission parameters (A'B'C'D').; Inter-parameter conversions and relationships..

Core References: Engineering Circuit Analysis — William H. Hayt & Jack E. Kemmerly (McGraw Hill), Network Analysis — M.E. Van Valkenburg (Prentice Hall)

SIGNALS & SYSTEMS (7 - 9 Marks)

- **Continuous-Time and Discrete-Time Signals:**

  Signal classifications: Continuous-time (CT) vs. Discrete-time (DT), deterministic vs. random, causal vs. non-causal.; Periodic vs. aperiodic signals: Fundamental period calculation for sum of continuous and discrete sinusoids.; Energy signals, Power signals, and neither energy nor power signals. Energy Spectral Density (ESD) and Power Spectral Density (PSD)..

- **Linear Time-Invariant (LTI) Systems:**

  System properties: Linearity (homogeneity and additivity), Time-invariance, Causality, Stability (BIBO stability criteria).; Memory (static vs. dynamic), Invertibility and inverse systems.; Impulse response h(t) and h[n]. Convolution integral and convolution sum: Properties (commutative, associative, distributive)..

- **Fourier Series and Continuous-Time Fourier Transform (CTFT):**

  Continuous-Time Fourier Series (CTFS): Trigonometric and Exponential Fourier series, Dirichlet's conditions.; Fourier coefficients, line spectrum, Parseval's power theorem for periodic signals.; Continuous-Time Fourier Transform (CTFT): Direct and inverse transform equations..

- **Sampling Theorem and Signal Reconstruction:**

  Sampling theorem for low-pass and band-pass signals. Nyquist rate and Nyquist interval.; Ideal impulse train sampling, flat-top sampling, and natural sampling.; Aliasing phenomenon and anti-aliasing filters. Guard bands..

- **Laplace Transform and Z-Transform:**

  Laplace Transform: Bilateral and unilateral transforms, Region of Convergence (ROC) properties.; Poles and zeros, stability and causality conditions in terms of ROC (ROC includes jw axis for stability; ROC right of rightmost pole for causality).; Properties of Laplace transform, initial and final value theorems, inverse Laplace transform using partial fractions..

- **Discrete-Time Fourier Transform & FFT Basics:**

  Discrete-Time Fourier Transform (DTFT): Periodicity of DTFT (2*pi periodic), properties, symmetry relations.; Discrete Fourier Transform (DFT): N-point DFT and IDFT definitions, circular shift, circular convolution vs. linear convolution.; Fast Fourier Transform (FFT): Decimation-in-Time (DIT) and Decimation-in-Frequency (DIF) radix-2 FFT algorithms, computational complexity reduction (O(N log2 N) vs O(N^2))..

Core References: Signals and Systems — Alan V. Oppenheim, Alan S. Willsky & S. Hamid Nawab (Pearson), Signals and Systems — Simon Haykin & Barry Van Veen (Wiley)

ELECTRONIC DEVICES (EDC) (9 - 11 Marks)

- **Semiconductor Physics & Carrier Transport:**

  Energy bands in intrinsic and extrinsic semiconductors: Conduction band, valence band, bandgap energy (Eg for Si = 1.12 eV, Ge = 0.66 eV, GaAs = 1.42 eV).; Direct vs. indirect bandgap semiconductors, electron and hole effective masses.; Equilibrium carrier concentrations: Fermi-Dirac distribution function, Fermi level position, intrinsic carrier concentration (ni)..

- **P-N Junctions, Zener & Special Diodes:**

  P-N junction in thermal equilibrium: Built-in potential (Vbi = Vt * ln(Na*Nd / ni^2)), energy band diagram across junction.; Depletion region under forward and reverse bias: Space charge density, electric field distribution (triangular for step junction, maximum at metallurgical junction), depletion width (W proportional to sqrt(Vbi - V)).; Junction capacitances: Depletion (transition) capacitance (dominant in reverse bias) and Diffusion capacitance (dominant in forward bias due to stored minority carriers)..

- **Bipolar Junction Transistor (BJT):**

  BJT physical structure and operating principles: NPN and PNP doping profiles (Emitter heavily doped, Base narrow and lightly doped, Collector large area for heat dissipation).; Modes of operation: Forward active, Saturation, Cutoff, and Reverse active modes.; Current components in active BJT: Base transport factor (beta*), Emitter injection efficiency (gamma), common base current gain (alpha), common emitter current gain (beta = alpha / (1 - alpha))..

- **MOS Capacitor & MOSFET Physics:**

  MOS capacitor physics: Metal-Oxide-Semiconductor energy band diagram at equilibrium, work function difference (Phi_ms).; Three regimes of MOS capacitor operation: Accumulation, Depletion, and Inversion (weak and strong inversion).; Flat-band voltage (Vfb), Threshold voltage (VT) derivation: Surface potential (phi_s = 2 * phi_F at onset of strong inversion), oxide capacitance (Cox = epsilon_ox / tox)..

- **Integrated Circuit (IC) Fabrication Basics:**

  Crystal growth (Czochralski method) and silicon wafer preparation.; Oxidation: Dry oxidation (thin, high-density gate oxide) vs. Wet oxidation (thick field oxide, faster growth, Deal-Grove model).; Photolithography: Photoresists (positive and negative), photomasks, resolution limits..

Core References: Semiconductor Physics and Devices — Donald A. Neamen (McGraw Hill), Physics of Semiconductor Devices — S.M. Sze & Kwok K. Ng (Wiley)

ANALOG CIRCUITS (8 - 11 Marks)

- **Diode Circuits & Power Supplies:**

  Small-signal diode model: Dynamic resistance (rd = eta * Vt / ID), diffusion and depletion capacitance effects.; Clipping circuits: Single-ended and double-ended diode clippers with DC reference voltages.; Clamping circuits (DC restorers): Peak clampers, clamping theorem (area of forward current = area of reverse current)..

- **BJT and MOSFET Amplifiers & Biasing:**

  BJT biasing and DC operating point (Q-point): Fixed bias, collector-to-base feedback bias, self-bias (voltage divider bias).; Stability factors: S(Ico), S(Vbe), S(beta). Thermal runaway condition: dPc/dT < 1/theta.; MOSFET biasing circuits: Voltage divider bias, current source active biasing..

- **Differential, Multi-Stage & Feedback Amplifiers:**

  Multi-stage amplifiers: RC-coupled, direct-coupled, and transformer-coupled amplifiers. Overall gain, bandwidth shrinkage in cascade.; Differential pair amplifiers (BJT and MOS pairs): Large-signal characteristics, small-signal differential gain (Ad), common-mode gain (Acm).; Common Mode Rejection Ratio: CMRR = |Ad / Acm|, CMRR in dB. Input common-mode range (ICMR), differential input resistance..

- **Operational Amplifiers (Op-Amps) & Applications:**

  Ideal vs. practical operational amplifier characteristics: Open-loop gain (Aol -> infinity), input resistance (Rin -> infinity), output resistance (Rout -> 0), bandwidth, offset voltage, bias currents, offset current.; Slew Rate (SR = dVo/dt_max = 2*pi*f_max*Vom), Full-power bandwidth (fpbw).; Virtual ground and virtual short concepts in negative feedback configurations..

- **Active Filters, Oscillators & 555 Timers:**

  Active filters: First-order and second-order low-pass, high-pass, band-pass, and band-reject (notch) Butterworth filters. Sallen-Key topology.; Oscillator principles: Positive feedback, Barkhausen criterion for sustained oscillations (|A*beta| = 1 and phase shift = 0 or 360 degrees).; RC oscillators: RC phase-shift oscillator (f0 = 1 / (2*pi*R*C*sqrt(6)), BJT/Op-Amp gain requirement Av >= 29), Wien-bridge oscillator (f0 = 1 / (2*pi*R*C), gain requirement Av >= 3)..

Core References: Microelectronic Circuits — Adel S. Sedra & Kenneth C. Smith (Oxford University Press), Electronic Devices and Circuit Theory — Robert L. Boylestad & Louis Nashelsky

DIGITAL CIRCUITS (8 - 10 Marks)

- **Number Systems & Boolean Algebra:**

  Number systems: Binary, octal, decimal, hexadecimal. Inter-base conversion algorithms.; Signed number representations: Signed magnitude, 1's complement, and 2's complement representations. Range of n-bit signed integers (-2^(n-1) to +2^(n-1) - 1).; 2's complement arithmetic, addition, subtraction, arithmetic overflow conditions (overflow occurs when sign of operands differs from sign of result)..

- **Combinational Logic Circuits:**

  Arithmetic circuits: Half adder, Full adder, Half subtractor, Full subtractor. Ripple carry adder (propagation delay tp = n * t_gate), Carry look-ahead adder (CLA, fast addition, generate and propagate terms).; Binary parallel subtractors, BCD adder, magnitude comparators (1-bit and multi-bit).; Multiplexers (MUX): 2:1, 4:1, 8:1, 16:1 MUX architecture. Implementation of boolean functions using multiplexers with and without external gates (Shannon's expansion)..

- **Sequential Logic Circuits:**

  Latches vs. Flip-Flops: Level-triggered vs. edge-triggered storage elements.; Basic SR latch (NAND and NOR implementations), race conditions, gated SR latch.; Flip-flops: JK flip-flop, Race-around condition in level-triggered JK flip-flop (occurs when pulse width tp > propagation delay tff). Solutions: Master-Slave JK flip-flop, edge triggering..

- **Finite State Machines (FSM) & Programmable Devices:**

  Synchronous sequential circuit design methodology: State diagram, state table, state reduction (equivalent states), state assignment.; Mealy vs. Moore state machines: Mealy machine (output depends on present state AND current input, faster response, fewer states), Moore machine (output depends ONLY on present state, glitch-free output).; Sequence detectors: Overlapping vs. non-overlapping sequence detection (e.g., detecting '1011' bit patterns)..

- **Data Converters & Semiconductor Memories:**

  Digital-to-Analog Converters (DAC): Binary weighted resistor DAC, R-2R ladder DAC (avoids wide range of resistor values, inverted ladder topology).; DAC specifications: Resolution (V_LSB = V_ref / (2^n - 1)), accuracy, linearity, settling time, monotonic behavior.; Analog-to-Digital Converters (ADC): Flash (parallel comparator) ADC (fastest, requires 2^n - 1 comparators), Successive Approximation Register (SAR) ADC (requires n clock cycles, binary search algorithm), Dual-slope integrating ADC (slowest, highest accuracy, immune to power supply fluctuations and 50 Hz noise), Counter-ramp ADC..

- **Microprocessor (8085) Architecture & Programming:**

  8085 Microprocessor internal architecture: ALU, accumulator (Reg A), general purpose registers (B, C, D, E, H, L), program counter (PC, 16-bit), stack pointer (SP, 16-bit), temporary registers.; Flag register: Sign (S), Zero (Z), Auxiliary Carry (AC), Parity (P), Carry (CY).; Address and Data buses: Multiplexed address/data bus (AD0 - AD7) and demultiplexing using ALE (Address Latch Enable) and 74LS373 latch..

Core References: Digital Design — M. Morris Mano & Michael D. Ciletti (Pearson), Fundamental of Digital Circuits — A. Anand Kumar (PHI)

CONTROL SYSTEMS (8 - 10 Marks)

- **Basic Control System Concepts & Modeling:**

  Open-loop vs. closed-loop control systems: Feedback effects (sensitivity reduction, bandwidth expansion, stabilization).; Transfer function concept: Definition (Laplace transform of impulse response with zero initial conditions), poles, zeros, proper and strictly proper transfer functions.; Block diagram representation and reduction rules: Moving summing points, moving takeoff points, parallel and feedback blocks..

- **Time-Domain Analysis & Transient Response:**

  Standard test signals: Unit step, unit ramp, unit parabolic, unit impulse.; First-order system response to step, impulse, and ramp inputs: Time constant (tau), settling time.; Second-order system transient response: Standard characteristic equation s^2 + 2*zeta*omega_n*s + omega_n^2 = 0..

- **Stability & Routh-Hurwitz Criterion:**

  Concept of stability in linear systems: Absolute stability vs. relative stability, BIBO stability criteria.; Location of poles in s-plane: Left half s-plane (stable), Right half s-plane (unstable), jw-axis poles (marginally stable if simple, unstable if multiple).; Routh-Hurwitz Stability Criterion: Necessary condition (all coefficients present and same sign), sufficient condition (first column of Routh array contains no sign changes)..

- **Root Locus Technique:**

  Root locus concept: Trajectory of closed-loop poles as open-loop gain K varies from 0 to infinity.; Angle condition: sum(angle of zeros) - sum(angle of poles) = +-(2q + 1)*180 degrees (determines whether a point in s-plane lies on root locus).; Magnitude condition: |G(s)H(s)| = 1 (determines the value of K for a valid point on the root locus)..

- **Frequency-Domain Analysis (Bode & Nyquist):**

  Frequency response specifications: Resonant peak (Mr = 1 / (2*zeta*sqrt(1 - zeta^2))), Resonant frequency (omega_r = omega_n*sqrt(1 - 2*zeta^2)), Bandwidth (BW = omega_n * sqrt(1 - 2*zeta^2 + sqrt(4*zeta^4 - 4*zeta^2 + 2))).; Bode plot: Semi-logarithmic magnitude plot (dB vs. log omega) and phase plot (degrees vs. log omega).; Corner frequencies, slope contributions (+-20 dB/dec for simple zeros/poles, +-40 dB/dec for complex pairs)..

- **Compensators & State-Space Analysis:**

  Compensator design: Lead compensator (dominant zero, adds positive phase angle, increases bandwidth, improves transient speed, increases PM), Lag compensator (dominant pole, improves steady-state accuracy by increasing low-frequency gain, reduces bandwidth), Lag-Lead compensator.; Maximum phase lead angle: sin(phi_m) = (1 - alpha) / (1 + alpha) at frequency omega_m = 1 / (T*sqrt(alpha)).; State-space representation: State variables, state vector x(t), input vector u(t), output vector y(t). State equations: dx/dt = A*x + B*u, y = C*x + D*u..

Core References: Modern Control Engineering — Katsuhiko Ogata (Pearson), Control Systems Engineering — Norman S. Nise (Wiley)

COMMUNICATION SYSTEMS (9 - 12 Marks)

- **Random Signals, Processes & Noise:**

  Random processes: Continuous-time random process, ensemble averages, stationarity: Wide-Sense Stationary (WSS) and Strict-Sense Stationary (SSS).; Autocorrelation function (ACF) R_xx(tau): Properties (R_xx(0) = total average power, even symmetry R_xx(tau) = R_xx(-tau), maximum at origin |R_xx(tau)| <= R_xx(0)).; Cross-correlation function and cross-spectral density..

- **Analog Communications (Modulation & Demodulation):**

  Amplitude Modulation (AM): Standard AM (DSB-FC) time and frequency domain expression, modulation index (mu = Am / Ac).; Total transmitted power: Pt = Pc * (1 + mu^2 / 2), transmission efficiency (eta = mu^2 / (2 + mu^2), max efficiency = 33.3% for single tone sinusoidal modulation).; Double Sideband Suppressed Carrier (DSB-SC): Time and frequency domain expressions, 100% power efficiency, bandwidth = 2 * fm. Coherent demodulation and Costas loop..

- **Information Theory & Source Coding:**

  Measure of information: Self-information I(xi) = log2(1 / P(xi)) bits.; Entropy of a discrete memoryless source: H(X) = -sum(P(xi) * log2 P(xi)) in bits/symbol. Maximum entropy for M equiprobable symbols: H_max = log2(M).; Entropy of binary source: H(p) = -p*log2(p) - (1-p)*log2(1-p)..

- **Digital Baseband Transmission & Pulse Shaping:**

  Pulse Code Modulation (PCM) transmitter and receiver: Sampling, Uniform Quantization, Encoding.; Quantization step size: delta = (V_max - V_min) / L where L = 2^n is the number of quantization levels.; Quantization noise power: Nq = delta^2 / 12 for uniform quantizer..

- **Digital Passband Modulation & Detection:**

  Geometric representation of signals: Gram-Schmidt orthogonalization procedure, signal space constellation diagram, basis functions.; Optimum receiver: Matched filter receiver (impulse response h(t) = s(T - t), maximizes output peak SNR = 2*E / N0), Correlation receiver.; Binary modulation schemes: Binary Phase Shift Keying (BPSK), Binary Frequency Shift Keying (BFSK, coherent and non-coherent), Binary Amplitude Shift Keying (BASK / OOK)..

- **Error Control Coding Basics:**

  Linear block codes: Generator matrix G of size k x n, Parity check matrix H of size (n-k) x n. Codeword c = m * G, syndrome vector s = r * H^T.; Minimum Hamming distance (d_min): Detects up to (d_min - 1) errors, corrects up to floor((d_min - 1) / 2) errors.; Standard Hamming codes: Single-error correcting (7, 4) code with d_min = 3..

Core References: Communication Systems — Simon Haykin & Michael Moher (Wiley), Modern Digital and Analog Communication Systems — B.P. Lathi & Zhi Ding (Oxford)

ELECTROMAGNETICS (EMFT) (8 - 10 Marks)

- **Electrostatics & Magnetostatics:**

  Coordinate systems: Cartesian (x, y, z), Cylindrical (rho, phi, z), and Spherical (r, theta, phi). Differential length, surface, and volume vector elements. Gradient, divergence, and curl in all three coordinates.; Divergence theorem (Gauss-Ostrogradsky theorem) and Stokes' theorem.; Coulomb's Law, Electric field intensity (E), field due to continuous charge distributions (line, surface, and volume charge distributions)..

- **Maxwell's Equations & Time-Varying Fields:**

  Faraday's Law of Electromagnetic Induction: Transformer EMF and Motional EMF. Differential form: curl E = -dB/dt, Integral form: closed line integral E.dl = -d(phi)/dt.; Inconsistency of Ampere's Law for time-varying fields: Equation of continuity (div J = -d(rho_v)/dt).; Maxwell's displacement current density: Jd = dD/dt..

- **Uniform Plane Waves & Wave Propagation:**

  Electromagnetic wave equations for E and H derived from Maxwell's curl equations (Helmholtz wave equations in phasor form).; Uniform plane wave propagation in lossless dielectric media: Propagation constant (gamma = alpha + j*beta = j*omega*sqrt(mu*epsilon)). Phase constant beta = omega*sqrt(mu*epsilon) = 2*pi / lambda.; Phase velocity (vp = omega / beta = 1 / sqrt(mu*epsilon) = c / sqrt(epsilon_r*mu_r)), Intrinsic impedance (eta = sqrt(mu / epsilon); for free space eta0 = 120*pi ~ 377 Ohms)..

- **Transmission Lines & Smith Chart:**

  Distributed parameter transmission line model: Resistance (R in Ohms/m), Inductance (L in H/m), Conductance (G in S/m), Capacitance (C in F/m).; Telegrapher's equations in time and frequency domains.; Propagation constant: gamma = sqrt((R + j*omega*L)*(G + j*omega*C)) = alpha + j*beta..

- **Waveguides & Resonant Cavities:**

  Rectangular waveguides: Physical structure, electromagnetic boundary conditions (tangential E = 0 on conducting walls).; Transverse Electric (TE_mn) and Transverse Magnetic (TM_mn) waveguide modes. Impossibility of TEM mode in hollow single-conductor waveguides.; Cutoff frequency formula for TE_mn and TM_mn modes: fc_mn = (c / 2) * sqrt((m / a)^2 + (n / b)^2) where a is width and b is height (a > b)..

- **Antennas & Radiation:**

  Radiation mechanism: Accelerated charges, retarded vector potential A(r, t).; Infinitesimal dipole (Hertzian dipole, length dl << lambda): Radiation fields (E_theta and H_phi in far-field proportional to 1/r), Radiation resistance R_rad = 80*pi^2*(dl / lambda)^2.; Half-wave dipole antenna (length l = lambda / 2): Current distribution (sinusoidal, zero at ends, maximum at center), radiation resistance R_rad = 73 Ohms, input impedance ~ 73 + j42.5 Ohms..

Core References: Elements of Electromagnetics — Matthew N.O. Sadiku (Oxford University Press), Electromagnetic Waves and Radiating Systems — Edward C. Jordan & Keith G. Balmain

GENERAL APTITUDE (15 Marks Fixed)

- **Verbal Aptitude:**

  Basic English Grammar: Tenses, subject-verb agreement, noun-pronoun agreement, active and passive voice, direct and indirect speech.; Parts of speech: Articles, prepositions, conjunctions, conditional clauses, modal verbs.; Vocabulary in context: Synonyms, antonyms, homonyms, phrasal verbs, frequently confused words..

- **Quantitative Aptitude:**

  Data Interpretation: Data tables, bar charts, pie charts, line graphs, two- and three-dimensional plots, Venn diagrams.; Numerical computation and estimation: Ratios, proportions, percentages, profit and loss, simple and compound interest.; Powers, exponents, logarithms, surds, indices, factorization, linear and quadratic equations..

- **Analytical Aptitude:**

  Logic and deduction: Deductive and inductive reasoning, conditional logic (if-then statements, contrapositive, converse, inverse).; Syllogisms: Statements and conclusions, validity of categorical syllogisms using Venn diagrams.; Analogy and relationship identification: Word analogies, number analogies, symbol relationships..

- **Spatial Aptitude:**

  Transformation of 2D and 3D shapes: Translation, rotation (clockwise and counter-clockwise), scaling, reflection, mirroring.; Assembling and grouping shapes: Matching identical geometric shapes, combining 2D patterns into 3D objects.; Paper folding, punching, and cutting: Predicting unfolded paper patterns..

Core References: A Modern Approach to Verbal & Non-Verbal Reasoning — R.S. Aggarwal, Quantitative Aptitude for Competitive Examinations — R.S. Aggarwal